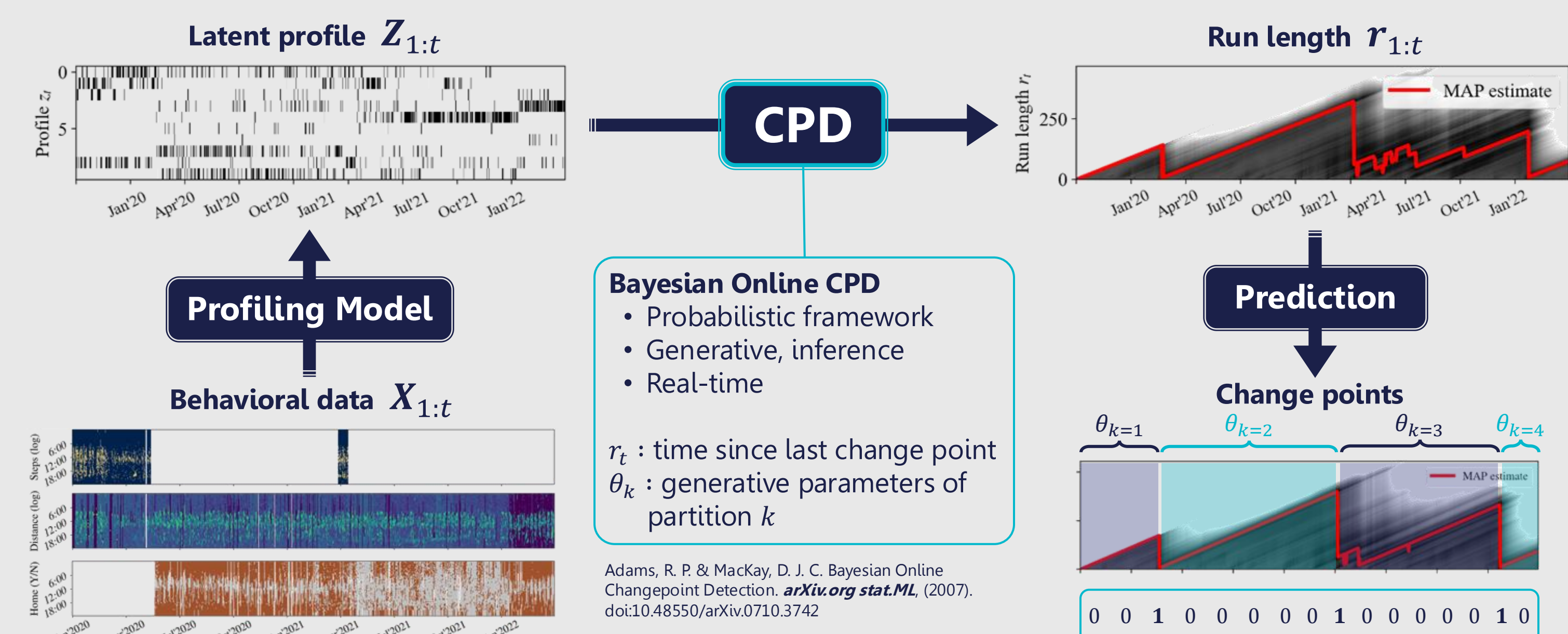


Detecting Behavioral Instabilities from Digital Phenotype

Change Point Detection



Inference in BOCPD

$$p(z_{t+1}|\mathbf{z}_{1:t}) = \sum_{r_t} p(z_{t+1}|r_t, z_t^{(r)}) p(r_t|\mathbf{z}_{1:t})$$

Run length posterior

$$p(r_t|\mathbf{z}_{1:t}) = \frac{p(r_t, \mathbf{z}_{1:t})}{p(\mathbf{z}_{1:t})} = \frac{\int p(r_t, \mathbf{z}_{1:t}, \theta_t) d\theta_t}{p(\mathbf{z}_{1:t})}$$

Joint distribution

$$p(r_t, \mathbf{z}_{1:t}, \theta_t) = \sum_{r_{t-1}} p(r_t, r_{t-1}, \mathbf{z}_{1:t}, \theta_t) = \sum_{r_{t-1}} p(r_t|r_{t-1}) p(\mathbf{z}_t|\theta_t) p(\theta_t|r_{t-1}, \mathbf{z}_{1:t-1}^{(r)}) p(r_{t-1}, \mathbf{z}_{1:t-1})$$

HetMM: Profiling

z_t : latent profile, takes value of 1 for the most probable out of K profiles, and 0 otherwise

Introduction of z_t in BOCPD

$$p(r_t, \mathbf{x}_{1:t}) = \iint p(r_t, \mathbf{z}_{1:t}, \mathbf{x}_{1:t}, \theta_t) d\theta_t d\mathbf{z}_{1:t}$$

Factorize $\mathbf{x}_{1:t}$ = $\iint p(\mathbf{x}_{1:t}|\mathbf{z}_{1:t}) p(r_t, \mathbf{z}_{1:t}, \theta_t) d\theta_t d\mathbf{z}_{1:t}$

Marginalize θ_t = $\int p(\mathbf{x}_{1:t}|\mathbf{z}_{1:t}) \int p(r_t, \mathbf{z}_{1:t}, \theta_t) d\theta_t d\mathbf{z}_{1:t}$ **Joint distribution in regular CPD**

$\phi_k^{(d)}$: parameters defining profile k in data type d

Joint likelihood

$$p(\mathbf{x}_t|z_t, \{\phi_k^{(1)}, \dots, \phi_k^{(D)}\}_{k=1}^K) = \prod_{k=1}^K \prod_{d=1}^D p(x_t^{(d)}|\phi_k^{(d)})^{\mathbb{I}\{z_t=k\}}$$

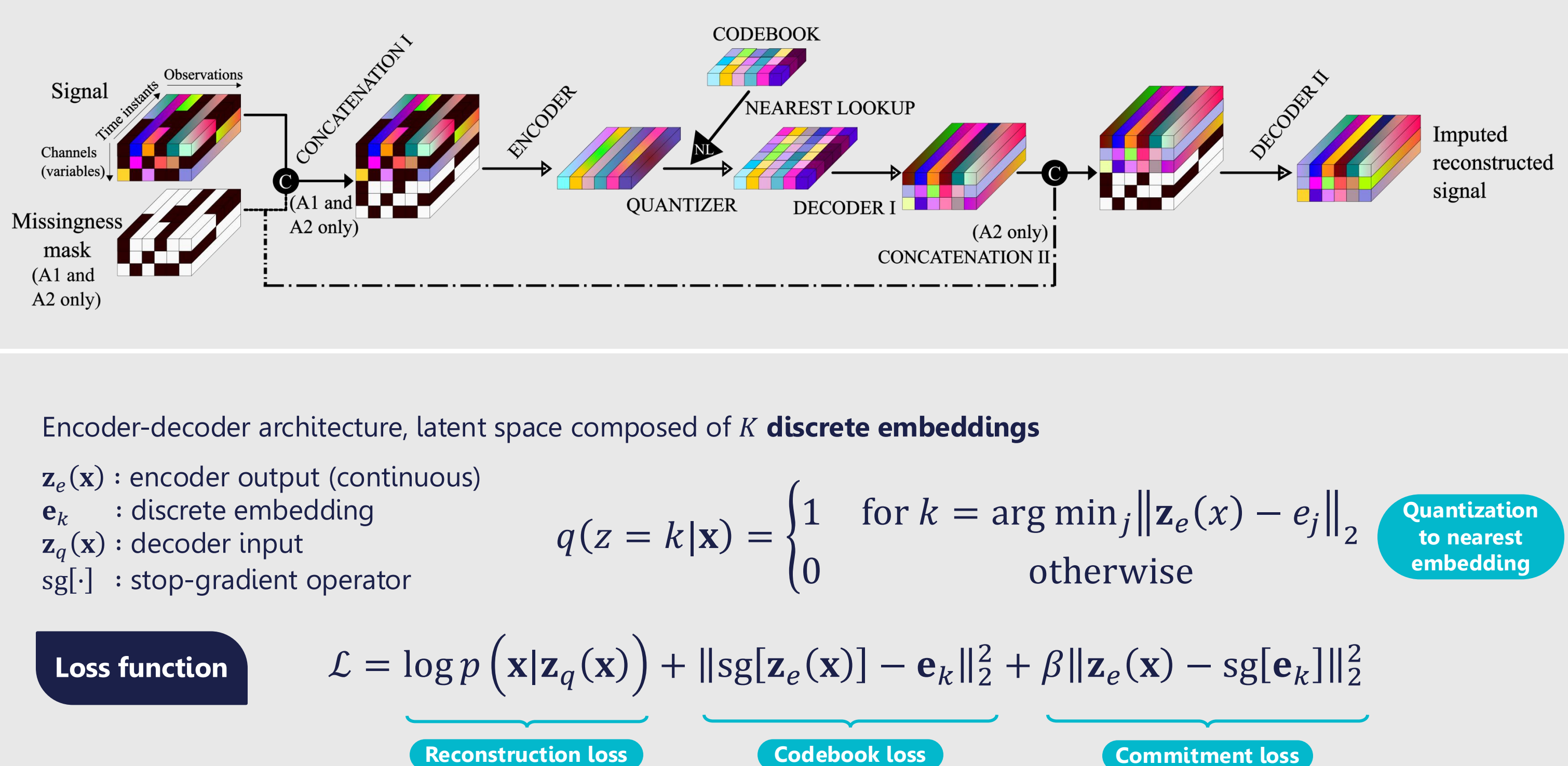
Complete likelihood

$$p(\mathbf{x}_t, z_t = k | \{\phi_k^{(d)}\}_{k=1, d=1}^{K,D}) = p(z_t = k) \mathbb{I}\{z_t=k\} \prod_{k=1}^K \prod_{d=1}^D p(x_t^{(d)}|\phi_k^{(d)})^{\mathbb{I}\{z_t=k\}}$$

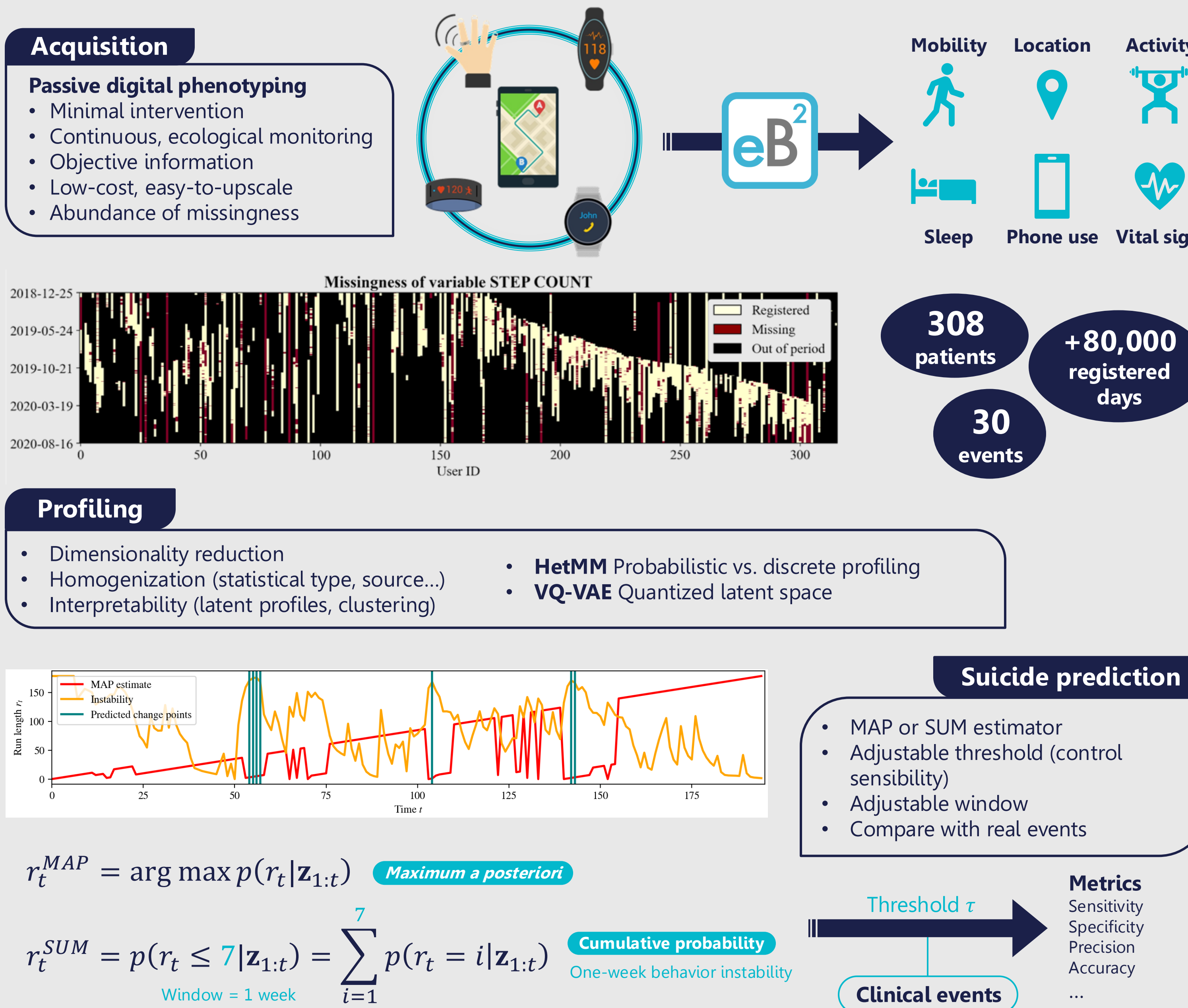
Loss function

$$\mathcal{L}_{z, \phi} = \log p(\mathbf{x}_{1:t}, \mathbf{z}_{1:t} | \{\phi_k^{(d)}, \pi_k\}_{k=1, d=1}^{K,D}) = \sum_{t=1}^T \sum_{k=1}^K \mathbb{I}\{z_t = k\} \log \pi_k + \sum_{t=1}^T \sum_{k=1}^K \sum_{d=1}^D \mathbb{I}\{z_t = k\} \log p(x_t^{(d)}|\phi_k^{(d)})$$

VQ-VAE: Discrete Embeddings



Short-term Suicidal Risk Prediction



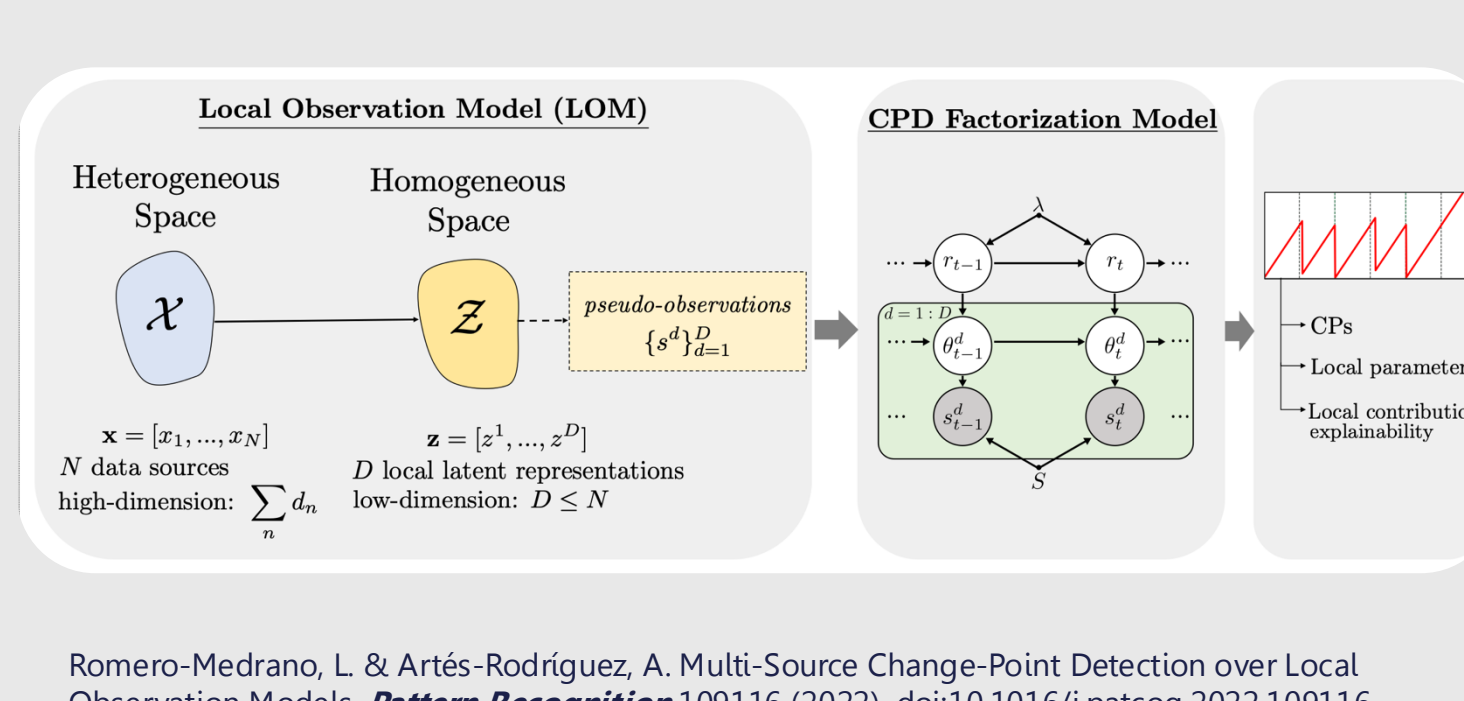
Multinomial sampling

$z_t^* = (z_t^{(1)}, \dots, z_t^{(S)}) \in \{1, \dots, K\}^S$ Multinomial profile

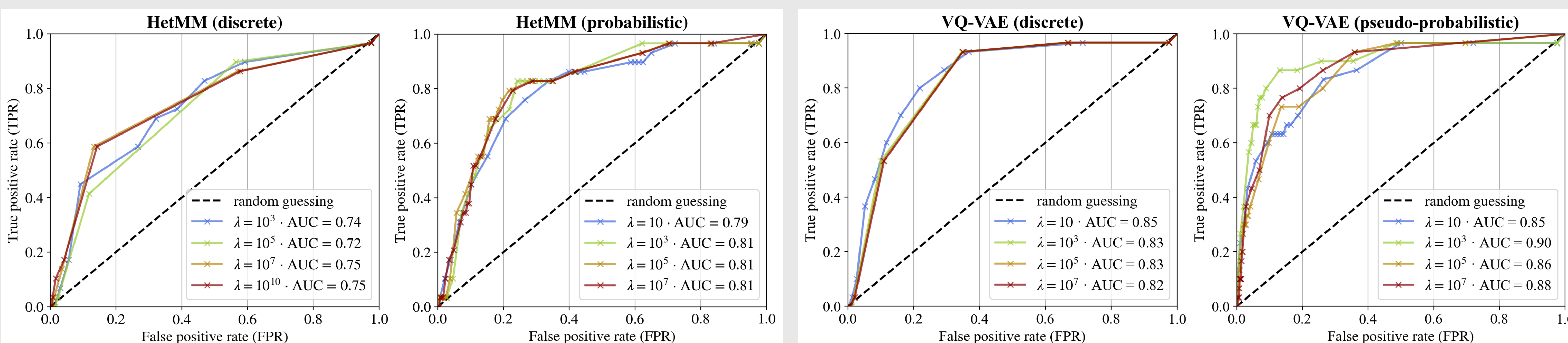
$c_t^k := \sum_{s=1}^S \mathbb{I}\{z_t^{(s)} = k\}$ **Counting vector**

$\theta_t \sim \text{Dirichlet}(\alpha)$ $c_t \sim \text{Multinomial}(\theta_t, S)$

Local observation models



Results



Heterogeneous Mixture Model

Vector-Quantized VAE